THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 29/09/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High fuel costs and extreme risks associated with live-jet pilot training.

2. AI RATIONALE & ANALYSIS

"Simulators accurately replicate physical flight dynamics and dangerous operational scenarios without consuming expensive fuel or risking equipment. The THEXRA platform provides an ideal deployment choice for high-fidelity, safe pilot instruction."

3. PROPOSED SOLUTION CONCEPT

Full-motion hardware flight simulator integrated with high-fidelity physics engines for safe, zero-fuel pilot training.

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

Simulator — THEXRA MECHALABO Heavy-Industry Training Simulators

1. Overview: A full-motion simulator environment powered by low-latency physics processing and immersive visual displays. The platform reproduces cockpit controls, atmospheric turbulence, and high-risk system failures in real time.

2. User Journey: Pilot boards motion cockpit → Instructor loads flight scenario → Pilot conducts mission → Real-time telemetry records performance → Post-flight debrief generated automatically.

3. Hardware: MECHALABO multi-axis dynamic motion flight rig, High-resolution panoramic visual projection screens, Replica jet avionics and control hardware

4. Software: MECHALABO Flight Physics Simulation Engine, Instructor Operator Station (IOS) Control Suite

5. AI Components: AI Atmospheric Turbulence and Aerodynamics Generator, Automated Flight Telemetry Analysis Module

6. XR Components: 180-degree Curved Visual Projection System, Spatial Flight HUD Overlay

7. Backend: Local Edge Compute Server for real-time physics calculation, High-speed UDP Telemetry Bus

8. Dashboard: Instructor Mission Control Dashboard, Pilot Flight Performance & Analytics Portal

9. Data Flow: Cockpit Controls → Local Edge Physics Engine → Motion Rig & Display → Instructor Telemetry Dashboard

Alternative Option: Combination solution — THEXRA Suite (Remote, Make, View)

5. EXPECTED BUSINESS OUTCOMES

Significant reduction in jet fuel expenses, elimination of live flight crash risks, and comprehensive telemetry tracking.

6. IMPLEMENTATION NEXT STEPS

Conduct a technical scoping session to review cockpit control integration and telemetry requirements.

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